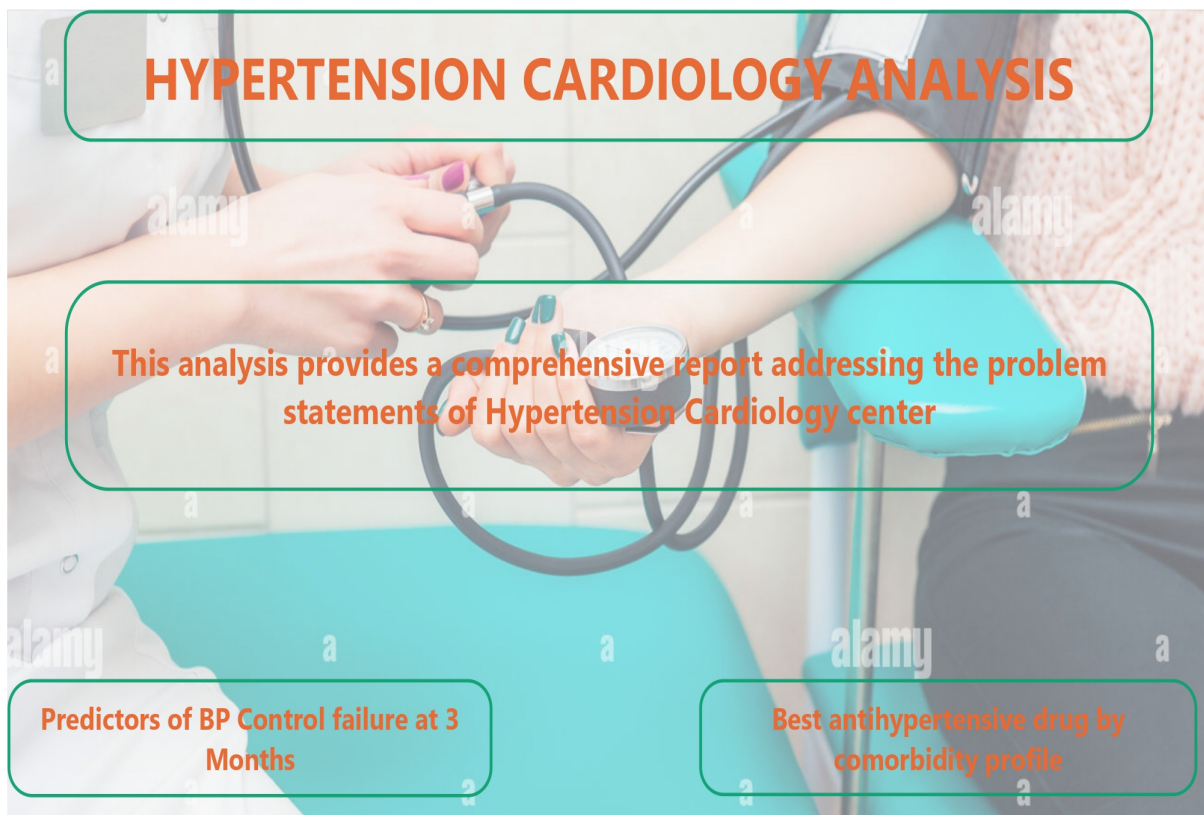


HYPERTENSION CARDIOLOGY CENTER

Clinical Performance & Treatment Optimization Report



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Role: Lead Clinical Data Analyst

Executive Summary

This analytical report details a data-driven evaluation of clinical data from a cohort of **350 patients** tracked over a 3-month timeline at the Hypertension Cardiology Center. The diagnostic population presents a care optimization hurdle: only **9.43% (33 patients)** reached fully controlled blood pressure thresholds, leaving an overall **90.57% (317 patients)** categorized as treatment control failures. This document delivers empirical insights addressing the foundational triggers of this failure and optimizes therapeutic choices across specific comorbidity cohorts.

Problem Statement 1: Point-of-Prescription Predictors of Failure

To isolate what drives 3-month blood pressure stabilization failure and systematically flag high-risk profiles at the initial consultation, a thorough evaluation of behavioral compliance and baseline indices was executed.

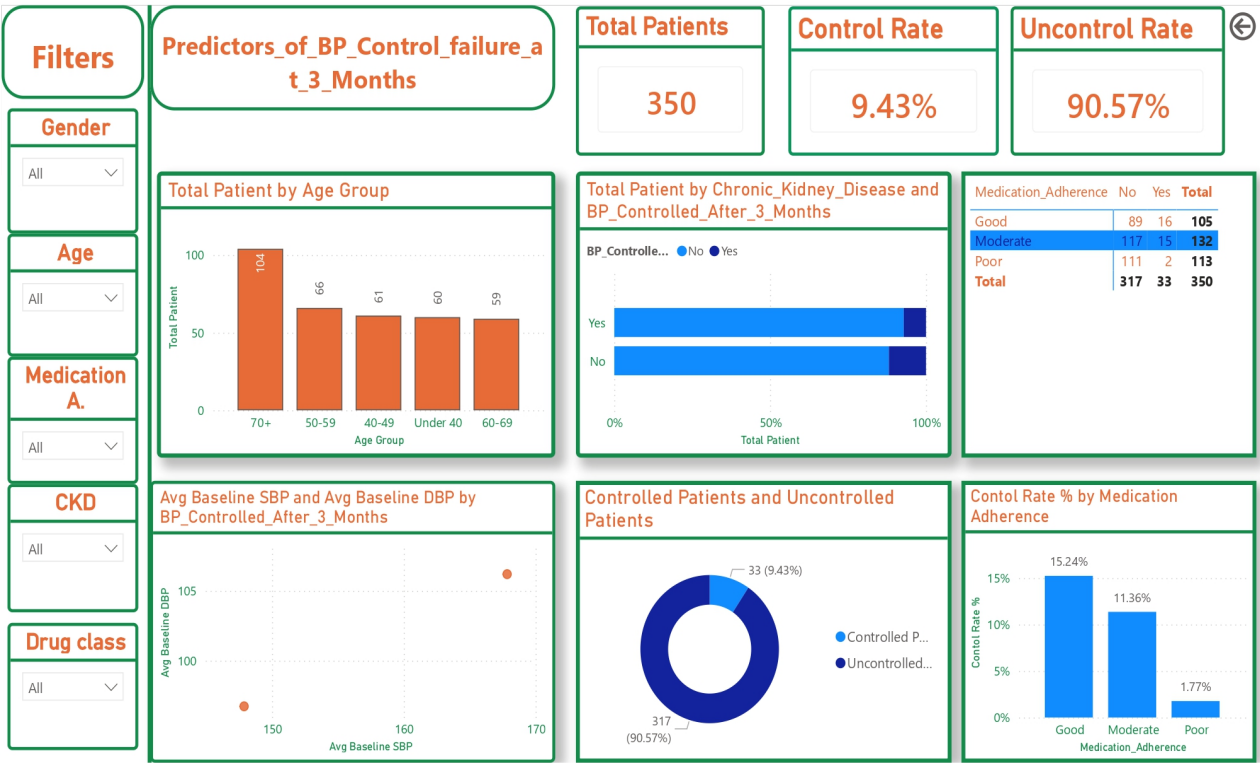


Figure 1: Risk Analysis Dashboard - Failure Metrics and Strategic Predictors.

Key Diagnostic Drivers:

- The Medication Adherence Factor:** Patient compliance serves as the most critical behavioral predictor of clinical failure. While patients displaying 'Good' adherence achieved a 15.24% control rate, those with 'Poor' compliance fell to a 1.77% control success rate (98.23% failure).
- Baseline Blood Pressure Severity:** The cohort enters treatment with advanced hypertensive baselines, averaging a Systolic BP of **165.96 mmHg** and a Diastolic BP of **105.31 mmHg**. Higher baseline values correlate tightly with long-term stabilization challenges.

- **Age Risk Layering:** The largest risk demographic resides within the **70+ senior cluster** (104 patients), where chronic conditions and physiological arterial stiffness lower responsiveness to regular single-drug interventions.

Problem Statement 2: Drug Class Efficacy & Comorbidity Optimization

Patients presenting with multi-layered comorbidities require highly specific therapeutics. By segmenting efficacy matrices across **Diabetes, Chronic Kidney Disease (CKD), Dyslipidemia, and Obesity**, distinctive performance gaps emerge.

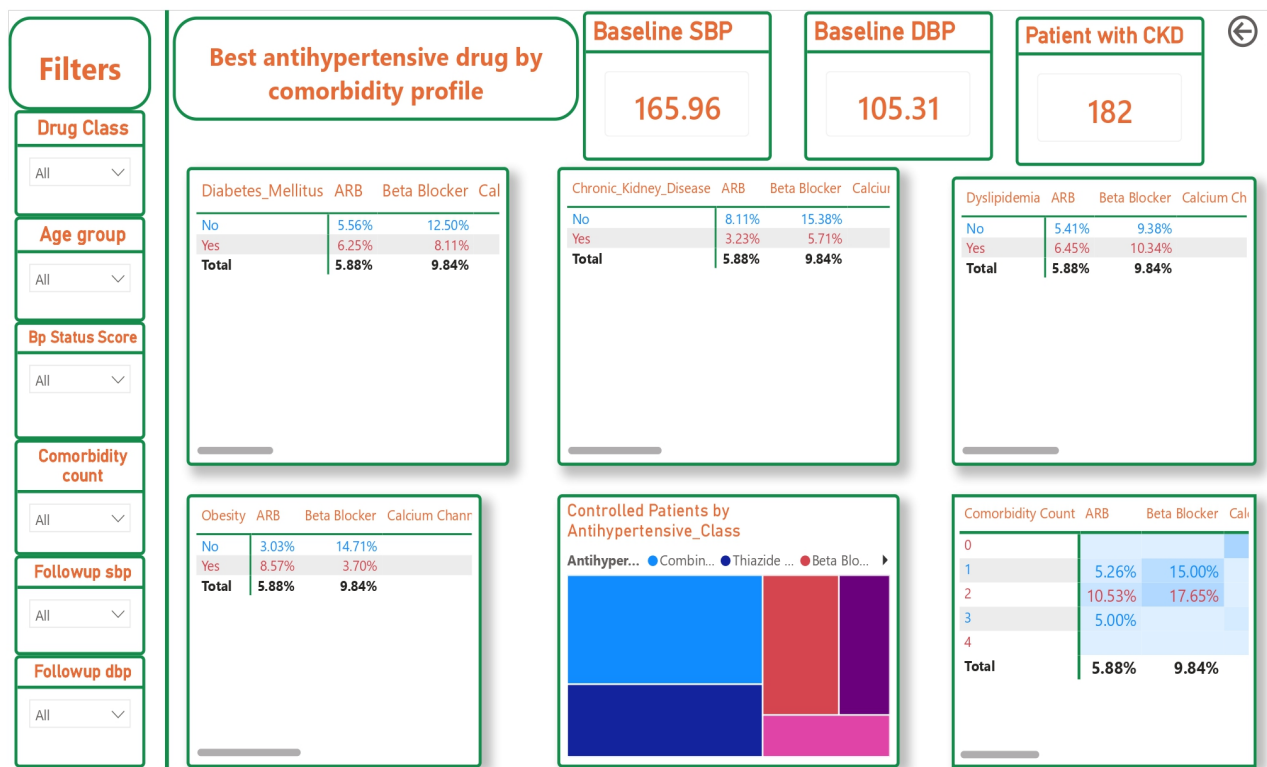


Figure 2: Therapeutic Optimization Matrix - Efficacy Mapping by Comorbidity Profiles.

Clinical Efficacy Insights:

- The Efficacy of Combination Therapy:** For diabetic profiles, Combination Therapy vastly outperformed alternative regimens, delivering an exceptional localized response (25.0% absolute control within its category vs. 0% for pure ACE Inhibitors).
- Chronic Kidney Disease (CKD) Stratification:** Out of 182 CKD patients, traditional monotherapies showed distinct limitations. Thiazide Diuretics demonstrated the highest numerical control response (6 controlled patients), while ACE Inhibitors yielded 0% control across all 27 CKD patients assigned.

Patient Comorbidity Profile	Optimal Drug Class Choice	Observed Clinical Note
Diabetes Mellitus	Combination Therapy	Highest success rate (8 Patients Controlled)
Chronic Kidney Disease	Thiazide Diuretics / Combination	ACE Inhibitors recorded a 0% response rate (0/27 Controlled)
Obesity & Multi-Comorbidity	Combination Therapy	Effectively targets high baseline systemic resistance

Strategic Recommendations & Operational Conclusion

1. **Point-of-Prescription Risk Flagging:** Integrate the logic derived in this study directly into the clinical workflow. Patients entering with Baseline SBP > 160 mmHg, advanced age (70+), and history of poor adherence must be flagged for intensive monitoring.
2. **Dynamic Adherence Interventions:** Implement automated SMS reminders and early-stage telehealth checks at Week 2 and Week 6 of the prescription lifecycle to mitigate adherence dropping.
3. **Protocol Upgrades for Complex Profiles:** Prioritize Combination Therapy configurations earlier in the cycle for multi-comorbidity or CKD patients to rapidly counteract extreme baseline pressures.